

UNIGE Mobile Robots exam - June 24, 2022 - 2 h

Work on paper, not on a tablet. Use a pen with good contrast (no pencil).

Exercise 1. Sketching a robot from parameter table (10 min)

Wheel	L	α	d	β	γ
1	a	π	0	β_1	0
2	a	$-\pi/2$	0	β_2	0
3	b	0	c	β_3	$-\pi/2$
4	b	$\pi/2$	c	β_4	$\pi/2$

Table 1.1. Parameters of the robot.

Using the table above, give the schematic representation of the robot. The type of the wheels is not indicated but can be inferred from the table. The figure should show vectors $x_1, \dots, x_4$, the dimensions a, b, c and the angles β_i . Vectors x_i and angles β_i will have the suffix f, s, c or g as appropriate, *e.g.* x_{1f} , β_{2c} . For β angles, make sure they are indicated by a circular arrow with the origin at the zero position of the angle. Frame R_m must be drawn. I recommend you first draw on scrap paper and **copy a clean result** on the answer sheet once you are done. Clarity and cleanliness of the figure will be evaluation criteria.

Exercise 2. Parametrization (10 min)

Parameterize the following robot (the lever arm b is identical for all wheels) and give the configuration vector.

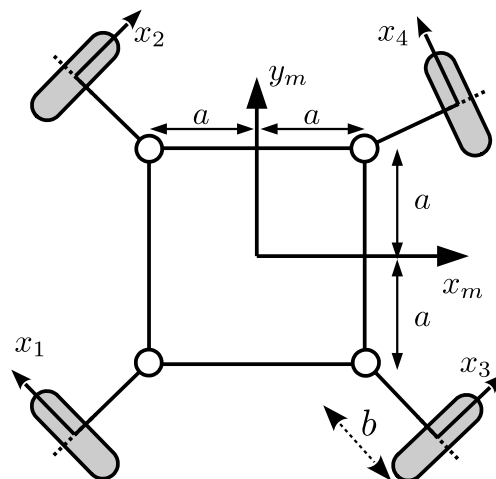


Figure 2.1. Robot to parametrize

Exercise 3. Kinematic modeling (30 min)

Wheel	L	α	d	β	γ
1s	a	0	0	β_{1s}	0
2c	b	$\pi/2$	c	β_{2c}	0
3c	b	$-\pi/2$	c	β_{3c}	0

Table 3.1. Parameters of the robot.

We want to establish the equations of the posture kinematic model of the robot of table 3.1.

- 1. Give the configuration vector.
- 2. Calculate matrices J_1 , J_2 , C_1 , C_2 and C_1^* .
- 3. What is the dimension of the kernel of C_1^* ?
- 4. Determine a base of the kernel of C_1^* . Make sure it is a base **whatever the value of β_{1s}** .
- 5 Calculate the posture kinematic model of the robot.

Exercise 4. Motorization study (20 min)

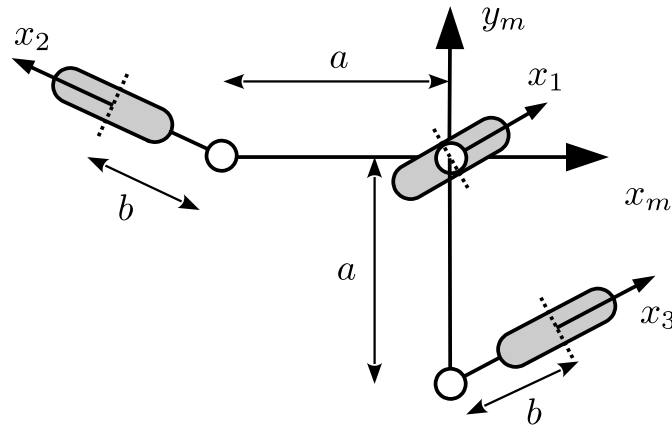


Fig. 4.1. Robot considered in this study

In this study, we consider the motorization of the robot of figure 4.1. In the equations below, the angles β are the standard angles used in the class material. Suffices s and c of the steerable and castor wheels have been omitted for simplicity.

The motorization matrix of the robot is given below:

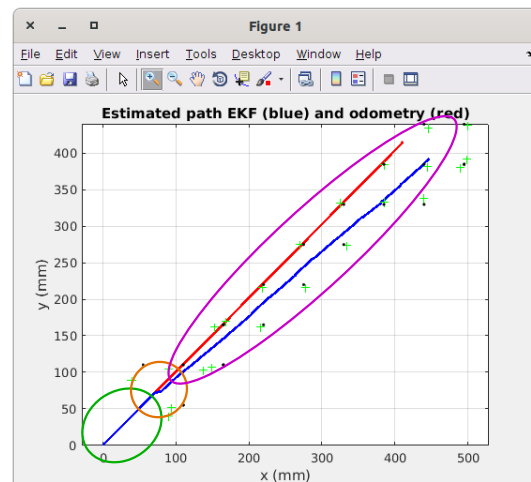
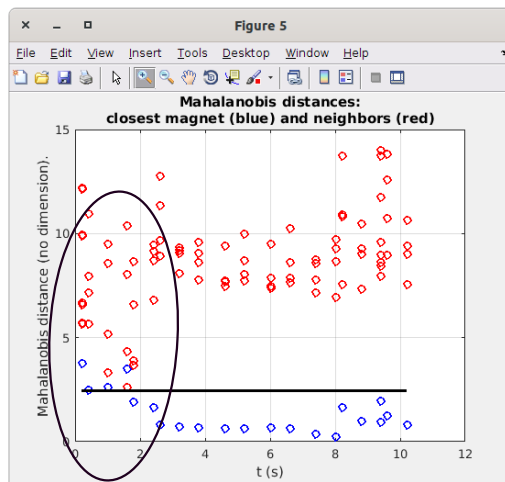
$$F = \begin{bmatrix} \frac{1}{b} \sin(\beta_1 - \beta_2) & -\frac{1}{b} (b + a \cos \beta_2) \\ -\frac{1}{b} \cos(\beta_1 - \beta_3) & -\frac{1}{b} (b + a \cos \beta_3) \\ \frac{1}{R} & 0 \\ -\frac{1}{R} \cos(\beta_1 - \beta_2) & \frac{a}{R} \sin \beta_2 \\ -\frac{1}{R} \sin(\beta_1 - \beta_3) & \frac{a}{R} \sin \beta_3 \end{bmatrix}$$

1. Is $b > a$ a reasonable design choice? Why? (3 lines of text maximum).
2. Give the configuration vector of the robot.
3. Indicate to which joint each line of matrix F corresponds.

4. Without calculations, give the type of the robot (no justification required).
5. In addition to the compulsory orientation motor of the steerable wheel, what is the theoretical minimum number of motors required (no justification is necessary)?
- 6.a. Prove that, when the robot geometric parameters are reasonable, any line other than line 3 of F can be equal to zero (I'm not saying all of them simultaneously). Give the reason for line 1 (2 is similar) and line 4 (5 is similar). This **can and must** be short and clear.
- b. Can any motorization that involves motorizing the joint corresponding to line 3 be singularity free with a minimum number of motors? Use the result of 6.a. There is no **need** for calculation to answer this question.

Exercise 5. Lab related questions (30 min)

1. In the lab, the dataset “twoLoops” has played an important role. Why? (10 lines maximum)
2. Assuming parameters P_{init} , Q_γ , the Mahalanobis threshold and Q_α are correctly set (to 0 for Q_α), explain the methodology to set σ_{tuning} . Precise which graphs should be used in the process. Do not reference figures by their number but by their content. Write your text as a manual for an operator. I encourage you to write it on scrap paper and copy to your exam sheet a clear and concise result. 20 lines of text maximum.
3. In the lab, the localization program was run at 5 Hz. As you know, data was available at 20 Hz. Now assuming you want to run the lab program at 20 Hz, which of the following parameters will you need to modify? If you think they should be modified, suggest a new value or at least explain whether their value should be increased or decreased. 10 lines of text maximum.
Parameters to consider: P_{init} , Q_γ , the Mahalanobis threshold, Q_α , σ_{tuning} .
4. Consider the sample results of the figures below.



Explain:

- The coincidence of the two paths in the green ellipse.
- The jump in the orange ellipse.
- The divergence between paths in the magenta ellipse.
- The meaning and positions of the red/blue dots in the black ellipse.

Exercise 6. Observability study (20 min)

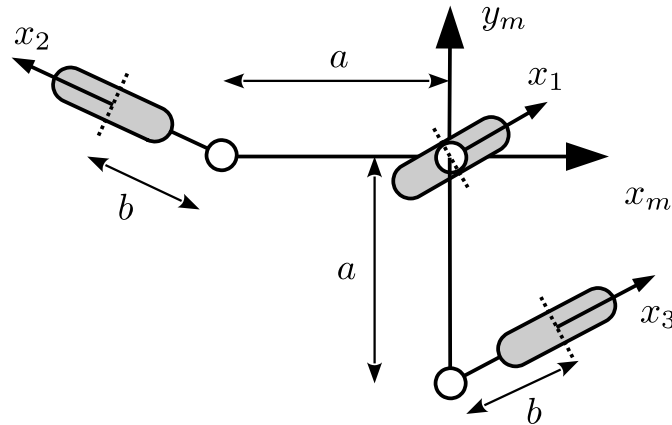


Figure 5.1. The robot

The robot of figure 5.1. is equipped with a GPS sensor, with the antenna at reference point M of the robot (center of the steerable wheel). Remember that a GPS measures the position of its antenna in the absolute reference frame R_0 .

With the standard notations of the class, the kinematics of the robot are given by the following equations:

$$\begin{cases} \dot{x} = u_1 \cos(\beta_1 + \theta) \\ \dot{y} = u_1 \sin(\beta_1 + \theta) \\ \dot{\theta} = u_2 \end{cases}$$

We want to study the observability of the posture of the robot.

1. Give a suitable state-space representation of the system for the observability study.
2. Calculate the gradients of the components of the measurement equation.
3. Calculate the first order Lie derivatives of the components of the measurement equation.
4. Calculate their gradients.
5. Using the previously calculated terms, define two sub-matrices of the observability matrix that consider the simplest to use in the observability study. Based on those two sub-matrices, find a simple condition under which the posture is observable.